A Appendix

A.1 Training the Learner Model

The architecture of the learner model is a Recurrent Neural Network (RNN) based on gated recurrent unit architecture [Cho et al., 2014]. It is implemented using Tensorflow and the gradients were calculated by automatic differentiation [Abadi et al., 2015]. At each step, the input contained the preceding learner action, represented as a one-hot vector, and the preceding reward, represented as a scalar. The action-reward history was shifted by one step, with an initial zero-padded input. Initial GRU hidden states were set to zero. The model returned both the GRU hidden states and a probability distribution over subsequent actions. The bandit model had two action outputs; the Multi-Round Trust Task model had five.

For MRTT, investments from 0 to 20 units were mapped to five discrete action categories: 0-4, 5-8, 9-12, 13-16, and 17-20 units. The learner networks were trained with Adam to minimise the summed negative log-likelihood of the observed action sequences (categorical cross-entropy for valid action observations). The implementation accommodated padded/missing actions when computing the sequence loss. During model development, 10-fold cross-validation was used for the bandit task and 5-fold cross-validation for MRTT to assess predictive performance and guide model and hyperparameter selection. Following this development stage, a final learner model for each LLM was trained on the full baseline dataset using the selected configuration. These final learner models were used in subsequent adversarial policy training. The optimal number of cells and iterations for each experiment are presented in Table 1.

Table 1: Optimal number of cells and training iterations

Experiment	**#cells in RNN**	**#training iterations**	**learning rate**
Bandit (GPT-3.5)	5	8600	0.005
Bandit (GPT-4)	5	1000	0.005
Bandit (Gemini-1.5)	5	2000	0.005
Bandit (DeepSeek-V3)	5	1300	0.005
MRTT (GPT-3.5)	3	3000	0.001
MRTT (GPT-4)	3	5000	0.001
MRTT (Gemini-1.5)	3	8000	0.001
MRTT (DeepSeek-V3)	3	6000	0.001

A.2 Training the Adversary

The Deep Q-learning algorithm was used for training the adversary in both the bandit task and the MRTT. In the bandit task, the reward for the adversary was 1 when the learner model selected the target action and 0 otherwise. In the MRTT, the reward for the MAX adversary in each trial was the earning amount ($3\times$ investment $-$ repayment), while in the case of the FAIR adversary, the reward was zero in each round except for the last round in which the reward was the negative absolute difference between the gains of trustee and investor over the whole task. The adversary neural network has three fully connected layers with 128, 128, and 4 units for the bandit task, and 128, 128, and 5 units for the MRTT, employing ReLU activation functions for the first two layers and a linear activation function for the final layer. Replay buffer sizes of 200,000 and 400,000 were considered. The ϵ-greedy method was used for exploration with $\epsilon \in \{0.01, 0.1, 0.2\}$. Learning rates 10^{-3}, 10^{-4}, and 10^{-5} were considered for training the adversary using the Adam optimizer. For policy evaluation and hyperparameter selection, each adversarial policy was evaluated over 200 simulations against the corresponding RNN learner model. The resulting average bias was used to select the hyperparameter configuration for each experiment. This evaluation was conducted separately from the subsequent adversarial evaluation involving the actual LLMs. The optimized combination of parameters for each experiment is shown in Table 2.

References

[Cho et al., 2014] Cho, Kyunghyun, Van Merriënboer, Bart, Gulcehre, Caglar, Bahdanau, Dzmitry, Bougares, Fethi, Schwenk, Holger, and Bengio, Yoshua. *Learning phrase representations using RNN encoder-decoder for statistical machine translation*. arXiv preprint arXiv:1406.1078, 2014.

[Abadi et al., 2015] Abadi, Martín, Agarwal, Ashish, Barham, Paul, Brevdo, Eugene, Chen, Zhifeng, Citro, Craig, Corrado, Greg S., Davis, Andy, Dean, Jeffrey, Devin, Matthieu, and others. *TensorFlow: Large-scale machine learning on heterogeneous systems*. Mountain View, CA: Tensorflow, 2015.

Table 2: The hyperparameters for training the adversary

Experiment	**buffer size**	**epsilon**	**learning rate**	**#training iterations**
Bandit (GPT-3.5)	400,000	0.01	10^{-3}	100,000
Bandit (GPT-4)	400,000	0.01	10^{-3}	100,000
Bandit (Gemini-1.5)	400,000	0.01	10^{-3}	100,000
Bandit (DeepSeek-V3)	400,000	0.01	10^{-3}	100,000
MRTT_MAX (GPT-3.5)	400,000	0.01	10^{-3}	40,000
MRTT_MAX (GPT-4)	400,000	0.01	10^{-3}	22,000
MRTT_MAX (Gemini-1.5)	400,000	0.01	10^{-3}	84,000
MRTT_MAX (DeepSeek-V3)	400,000	0.01	10^{-3}	36,000
MRTT_FAIR (GPT-3.5)	400,000	0.01	10^{-3}	42,000
MRTT_FAIR (GPT-4)	400,000	0.01	10^{-3}	70,000
MRTT_FAIR (Gemini-1.5)	400,000	0.01	10^{-3}	34,000
MRTT_FAIR (DeepSeek-V3)	400,000	0.2	1e-05	34,000